RESEARCH ARTICLE SUMMARY

MARINE CONSERVATION

Ecological erosion and expanding extinction risk of sharks and rays

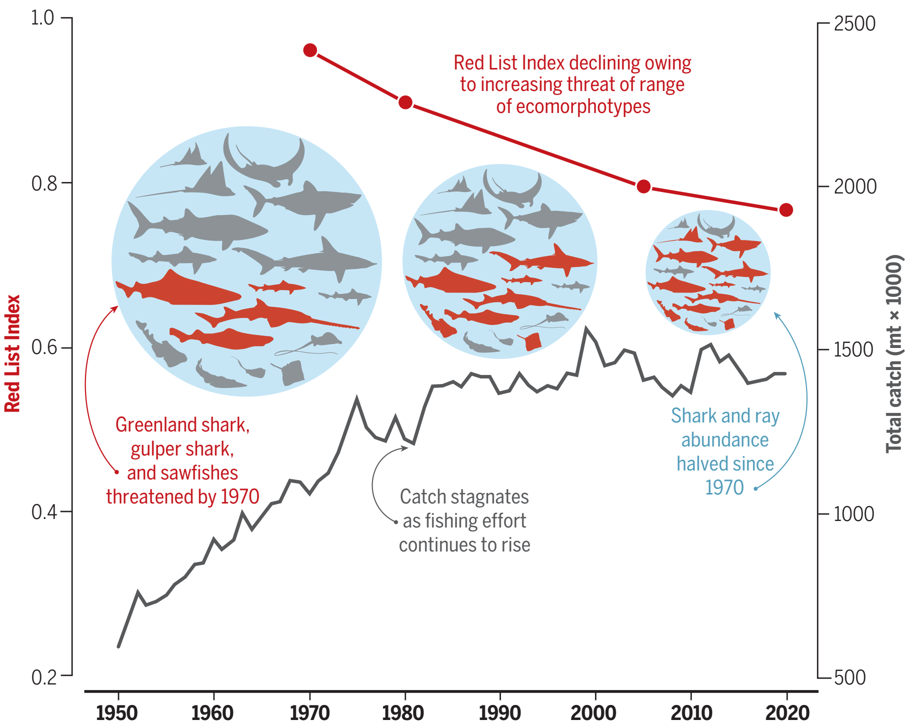
Nicholas K. Dulvy†*, Nathan Pacoureau†, Jay H. Matsushiba, Helen F. Yan, Wade J. VanderWright, Cassandra L. Rigby, Brittany Finucci, C. Samantha Sherman, Rima W. Jabado, John K. Carlson, Riley A. Pollom, Patricia Charvet, Caroline M. Pollock, Craig Hilton-Taylor, Colin A. Simpfendorfer

INTRODUCTION: The key to implementing global biodiversity targets and sustainable development goals is to hold nations accountable to their promises. This can be achieved by developing indicators to track progress (or lack thereof) toward these targets and goals. The Red List Index (RLI) is based on the International Union for Conservation of Nature (IUCN) Red List of Threatened Species assessments and is widely used to report on the status of terrestrial biodiversity and the effectiveness of conservation actions, yet there is no equivalent for the oceans. This indicator gap is particularly acute because both marine life and the key threat of overfishing are concealed and hard to observe.

RATIONALE: Chondrichthyans comprise at least 1199 predatory species found throughout the world's oceans. Many of these sharks and rays are intrinsically sensitive to overfishing and exhibit a range of important ecological functions, and it remains to be seen how their biodiversity and ecosystem function has diminished over the past half-century of fisheries expansion. By developing a 50-year RLI of sharks and rays, we have shown how extinction risk has expanded across geographic, bathymetric, and biodiversity axes, including the mapping of bright spots (of high richness with little change in RLI) and dark spots (of high richness with steep declines in RLI). In addition, we estimated ecological erosion in both the RLIs of ecomorphotypes and the decline of ecological function embodied in threatened species. Finally, we identified ecological, fisheries, and socioeconomic drivers of extinction risk (national range-weighted RLI).

RESULTS: Shark and ray catches initially rose with rising fishing efforts and then stagnated by the 1980s as populations collapsed, as revealed by the declining RLI. Since 1970, overfishing has halved shark populations, and the shark and ray RLI has worsened by 19%. This serial depletion began in rivers, estuaries, and nearshore coastal waters before expanding out across the oceans and then down into the deep sea. As a result, there has been an erosion of functionally important ecomorphotypes, and we estimate that the elimination of threatened species would result in a 22% reduction in functional diversity. National extinction risk was greater in countries with higher fishing pressure (lower fisheries catch rates and larger coastal populations) and with larger contributions of harmful fisheries subsidies. Nations with lower extinction risk had stronger governance, larger economies, and greater beneficial fisheries subsidies.

CONCLUSION: There has been notable progress in the appreciation and conservation of sharks and rays. Yet chronic underassessment and undermanagement of fisheries is widespread, particularly in countries with weaker governance. Science-based fisheries catch limits and measures to minimize incidental catch, including spatial protections, are essential to ensure sustainability and to recover species to their ecological, social, and economic potential. However, these limits need to be applied at scale, enforced, and tailored to species' biology. Finally, we recommend the adoption of the shark and ray RLI to track progress toward both Kunming-Montreal Global Biodiversity Framework target 5 and United Nations Sustainable Development Goal 14.4. ■



Global shark and ray RLI for tracking progress toward global biodiversity and sustainability targets. The global reconstructed catch of all chondrichthyans peaked in the late 1980s and has been stagnant ever since, despite a doubling of fishing effort. This is likely due to increasing unsustainability and population collapse, which has resulted in greater numbers of species at risk of extinction, as revealed by declines in the RLI since 1970. Over time, the RLI continued to decline owing to the serial depletion of the largest most functionally important sharks and rays, some to the point of local extinction (sawfishes and other rhino rays), before fisheries sequentially depleted large stingrays, eagle rays, angel sharks, hammerheads, and requiem sharks. As a result, there is an increasing representation of smaller individuals and species in catches. An RLI value of 1 indicates that a given taxa qualifies as IUCN Red List category Least Concern (that is, not expected to become extinct in the near future), whereas an RLI value of zero indicates that all taxa have gone extinct. mt, metric tons.

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Ecological erosion and expanding extinction risk of sharks and rays

Nicholas K. Dulvy[1]†*, Nathan Pacoureau[1,2,3]†, Jay H. Matsushiba[1], Helen F. Yan[4,5], Wade J. VanderWright[1], Cassandra L. Rigby[5], Brittany Finucci[6], C. Samantha Sherman[1,7], Rima W. Jabado[5,8,9], John K. Carlson[10], Riley A. Pollom[11], Patricia Charvet[12], Caroline M. Pollock[13], Craig Hilton-Taylor[14], Colin A. Simpfendorfer[5,14]

The true state of ocean biodiversity is difficult to assess, and there are few global indicators to track the primary threat of overfishing. We calculated a 50-year Red List Index of extinction risk and ecological function for 1199 sharks and rays and found that since 1970, overfishing has halved their populations and their Red List Index has worsened by 19%. Overfishing the largest species in nearshore and pelagic habitats risks loss of ecomorphotypes and a 5 to 22% erosion of functional diversity. Extinction risk is higher in countries with large human coastal populations but lower in nations with stronger governance, larger economies, and greater beneficial fisheries subsidies. Restricting fishing (including incidental catch) and trade to sustainable levels combined with prohibiting retention of highly threatened species can avert further depletion, widespread loss of population connectivity, and top-down predator control.

The unsustainable capture and trade of wild marine animals, such as sharks and rays, is a key threat to the marine biodiversity that underpins food security, livelihoods, sustainable development, and diverse maritime cultures (*1*, *2*). In 2022, 196 Parties to the United Nations (UN) Convention on Biological Diversity (CBD) committed to the Kunming-Montreal Global Biodiversity Framework (KMGBF), including target 5 to "ensure safe, sustainable and legal harvesting and trade of wild species." Similarly, the UN Sustainable Development Goals (SDGs) include target 14.4 on sustainable fishing and target 15.5 to "protect and prevent the extinction of threatened species" (on land) (*3*).

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Progress toward some of the biodiversity targets (KMGBF goal A and target 4) and SDGs (target 15.5) is tracked using the International Union for the Conservation of Nature's (IUCN) Red List Index (RLI), which is based on repeat assessments of extinction risk for the IUCN Red List of Threatened Species (*4*–*6*). Yet the tracking of sustainable fisheries (KMGBF target 5 and SDG target 14.4) relies on a mandated indicator that is based on population-level stock assessments of a few hundred fisheries (*6*). This index is geographically and taxonomically sparse and based mostly on coastal teleost fishes in North America, Europe, South Africa, Australia, and New Zealand and high-seas market tunas with little representation of small-scale fisheries or nontarget, incidentally caught species (*7*–*9*). There is a need for a global RLI to track trends in exploited aquatic biodiversity and the consequences for ecological function (*9*–*12*).

Sharks, rays, and chimaeras (class Chondrichthyes, hereafter "sharks and rays") are the oldest radiation of vertebrates, having arisen more than 420 million years ago (*13*), and they occur in a wide range of estuarine, marine, and even some freshwater habitats, dominating the higher trophic levels at the top of aquatic ecological pyramids (*10*, *12*). Sharks and rays exhibit the greatest diversity of vertebrate reproductive investment strategies, and most have slower intrinsic rates of population growth that are more similar to those of marine mammals than those of bony fishes, which renders them highly susceptible to overfishing (*14*). They are targeted or incidentally captured in multispecies fisheries, where they are discarded or retained as bycatch for local consumption and sale through domestic markets (*9*, *15*, *16*). Meat and other valuable body parts, from both target and incidental catch, are traded internationally, including meat, fins, skins, gill plates, and liver oil (*14*, *15*). Despite their importance to fisheries and ecosystems, as well as tourism potential for several species, sharks and rays are woefully undermanaged in most places (*1*, *17*, *18*). Shark and ray exploitation patterns in the face of low reproductive potential have markedly reduced many species' abundance, and a corresponding deterioration of survival probability and ecological functionality would not be surprising. In this work, we empirically assess whether the multiple dimensions of shark and ray biodiversity and ecosystem function are declining.

Specifically, we calculated an aquatic RLI for the world's 1199 sharks and rays spanning more than 50 years (1970–2020) based on one of the most comprehensive Red List reassessments authored by 322 assessors and 363 contributors and checked by 118 reviewers in consultation with more than 180 members of the IUCN Species Survival Commission Shark Specialist Group (*19*, *20*). Second, we compared the shark and ray RLI status and trends to other comprehensively assessed vertebrate and marine lineages. Third, we quantified the worldwide expansion of extinction risk along geographic, bathymetric, and biodiversity axes using subsets of the RLI. Fourth, we estimated ecological erosion as the decline of functional richness embodied in threatened species (IUCN Red List categories of Vulnerable, Endangered, and Critically Endangered). Fifth, we mapped the bright spots with high richness and low change in RLI compared with the dark spots of steep rarity-weighted RLI decline in places with high species richness. Finally, we identified the ecological, fisheries, and socioeconomic drivers of RLI within exclusive economic zones (EEZs) to guide management recommendations for reversing the decline of sharks and rays.

Fifty-year deterioration in the status of sharks and rays

The global reconstructed catch of all sharks and rays doubled from ~750,000 to ~1.5 million metric tons (mt) and fishing effort more than tripled between 1950 and 2000 (inset of Fig. 1). Consequently, there has been a 53.2% [95% credible interval (CI) 50.1, 56.2] reduction in relative abundance as indexed by catch per unit effort (CPUE) over half a century (1970–2019; Fig. 1). The overall decline is 64.8% (CI 61.4, 67.8) since 1951, which is still conservative compared with the estimated 81 to 89% decline of a group of 112 shark fisheries over the period 1975 to 2009 (*21*). These CPUE declines, along with other population trends (*22*, *23*), were used to assess the IUCN Red List status through the population reduction "A" criterion for 95.4% (n = 373) of threatened species (*20*).

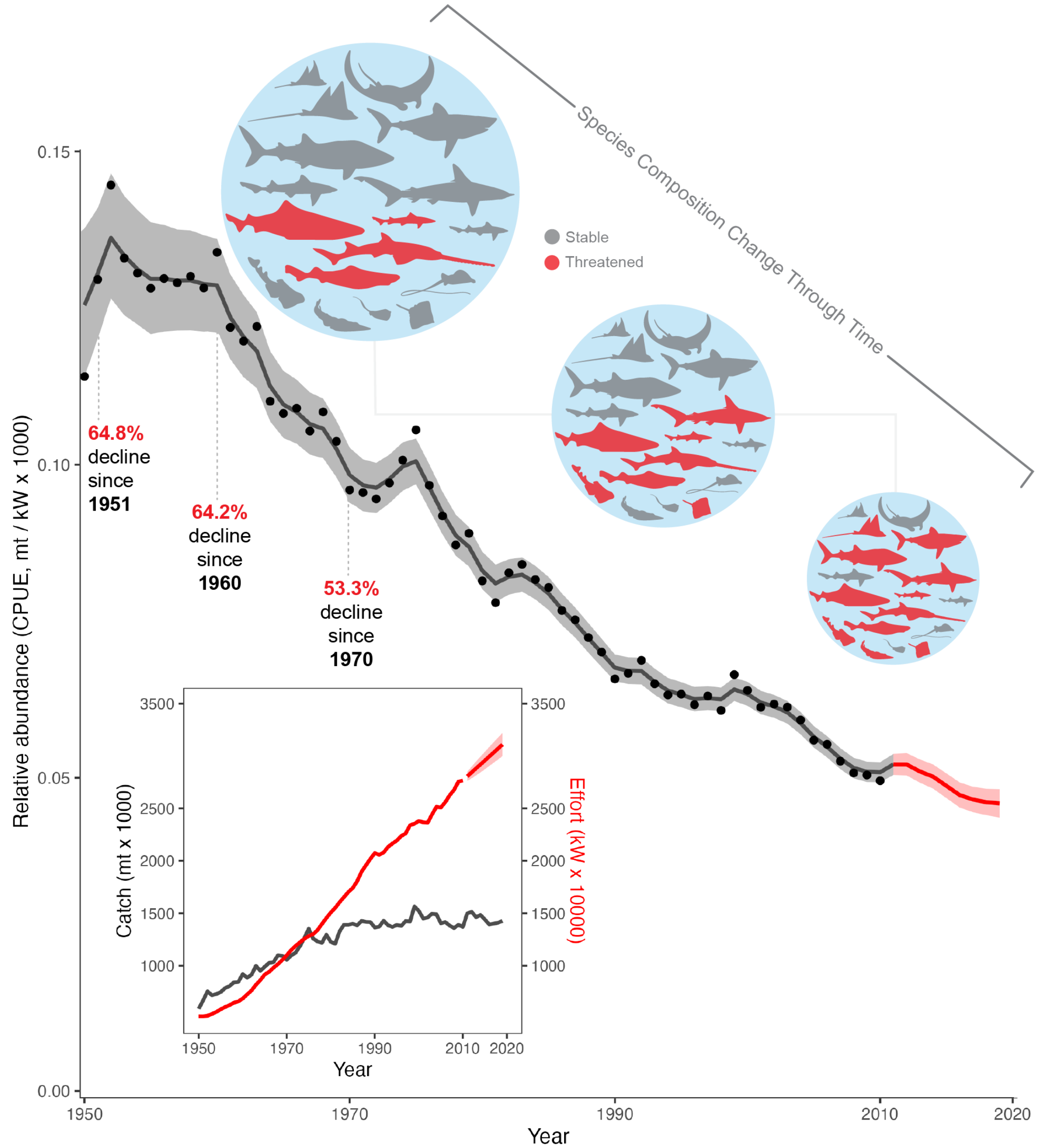

Fig. 1. Global decline of sharks and rays. Relative abundance of sharks and rays indexed by CPUE (in mt 1000 kW^{-1}) from 1950 to 2010, showing estimates of declines starting from 1951, 1960, and 1970. The inset shows stagnating global shark and ray catch (mt × 1000 within EEZs; black line) and rapidly rising global fishing effort (10,000 kilowatt hours, 10,000 kW^{-1} within and outside of EEZs; red line) from 1950 to 2010 with catch extending to 2019. The diagram above the CPUE line reflects how the declines in abundance (CPUE) were accompanied by sequential depletion and increasing threat of larger macropredatory ecomorphotypes, leaving relatively smaller mesopredatory ecomorphotypes; see Figs. 2 and 3. The shaded regions indicate 95% CIs.

At the same time, the global RLI of 1199 sharks and rays worsened by 19%; already depleted by 4% in 1970 (RLI = 0.96), the RLI declined to 0.77 by 2020 (Fig. 2A). The RLI was highly robust to uncertainty owing to Data Deficient species and classification error by assessors, with the RLI more likely to be worse if the true category of Data Deficient species was known (fig. S1A) and the classification uncertainty of the RLI within each year much smaller than the change in RLI over time (fig. S1, B and C). Shark and ray extinction risk worsened faster than that of all other comprehensively assessed vertebrate groups, apart from amphibians (*24*, *25*), and they are now more threatened than marine and freshwater fishes (assessed with the sampled RLI) and four comprehensively assessed marine groups: hagfishes, groupers, cone snails, and stony corals (*26*–*30*) (Fig. 2A).

By 1970, 3.5% (n = 37 of 1044) data-sufficient species were already threatened, with four species inferred to have been Critically Endangered (Pondicherry shark *Carcharhinus hemiodon* and barndoor skate *Dipturus laevis*) or Critically Endangered (possibly extinct) (lost shark *Carcharhinus obsoletus* and Java stingaree *Urolophus javanicus*), six Endangered, and 27 Vulnerable (for the category changes in the 1044 data-sufficient species, see Fig. 2B). By 1980, 15% (n = 157) of species were threatened, and by 2005, nearly one-third were threatened (32.5%, n = 339), consistent with our trait-based category prediction at the time (*19*). By 2020, 37.5% (n = 391) of data-sufficient species were assessed as threatened, and a total of 33.3 to 37.5% of all species were predicted to be threatened, assuming 155 Data Deficient species are threatened either in the same proportion as non–Data Deficient species or using a trait-based prediction of category (*20*) (Fig. 2B). Although the decline of the whiskery shark (*Furgaleus macki*) had already been halted by 2000, since then only three species have genuinely improved: the barndoor skate, smooth skate (*Malacoraja senta*), and New Zealand smooth skate (*Dipturus innominatus*) (fig. S2).

Sequential depletion of the largest, most sensitive species

The increasing extinction risk over time exhibits size-based serial depletion, with the largest species declining first and most steeply. Rays are 4% more threatened than sharks [RLI_{rays} = 0.74 and RLI_{sharks} = 0.78, i.e., a difference in RLI (ΔRLI) of 0.04; Fig. 3A]. The RLI of rays declines consistently across all size classes, with the larger-bodied rays declining faster than

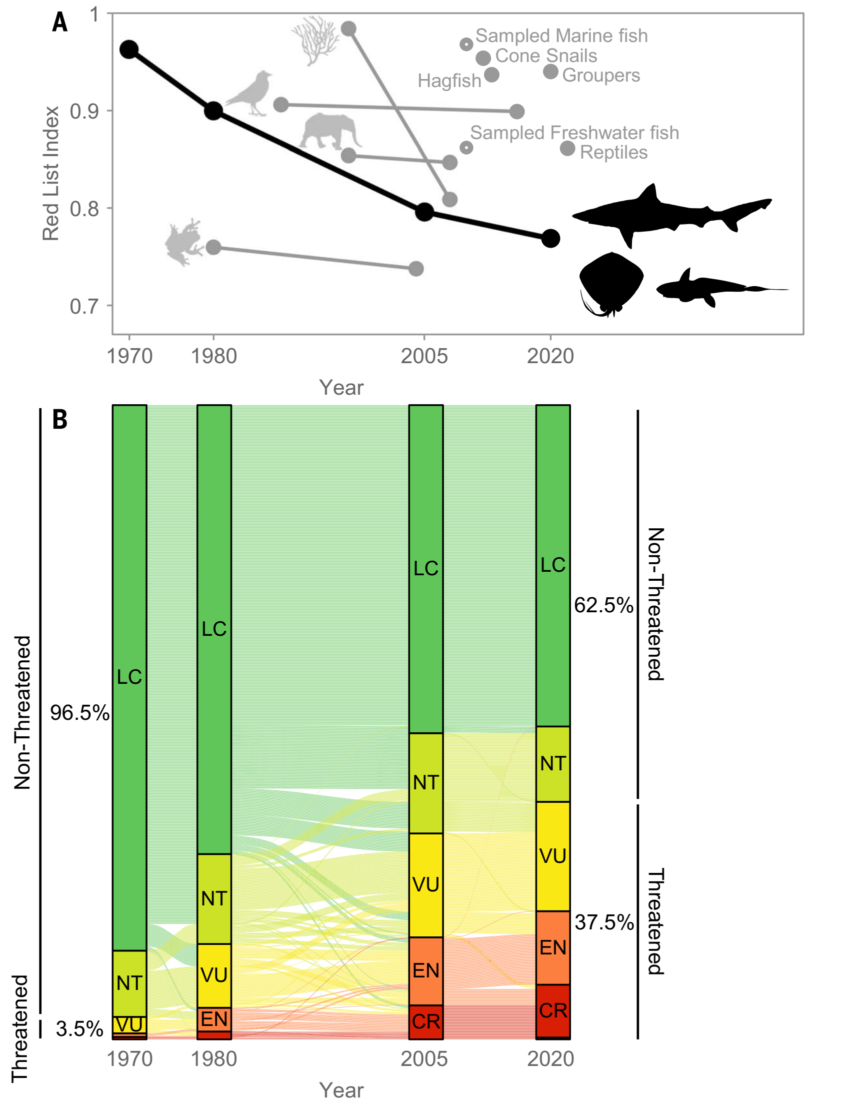

Fig. 2. Half a century of global biodiversity deterioration for comprehensively assessed groups from 1970 to 2020 as measured by the RLI. (**A**) A steep decline in the global RLI of sharks and rays (black line) inferred in 1970, 1980, 2005, and 2020 as well as RLIs of completely assessed groups (gray lines and solid dots), including mammals, birds, amphibians, reptiles, groupers, hagfishes, cone snails, and stony corals, and the sampled RLI estimates (hollow dots) for marine and freshwater fishes (*24*–*30*). A RLI value of 1.0 indicates that all species are Least Concern (i.e., do not have a high risk of extinction in the wild in the medium-term future), whereas a RLI value of 0 indicates that all species have gone extinct. (**B**) Change in the Red List category of the data-sufficient sharks and rays from 1970 to 2020 (n = 1044; i.e., excluding the 155 Data Deficient species). IUCN Red List categories are, from top to bottom (green to red), as follows: LC, Least Concern (dark green); NT, Near Threatened (light green); VU, Vulnerable (yellow); EN, Endangered (orange); and CR, Critically Endangered (red).

medium- and small-bodied rays. The RLI_{1970} of 0.90 for the 66 largest ray species was slightly worse in 1970 and declined more steeply over half a century to an RLI_{2020} of 0.61 than that of the medium-sized rays (n = 202; RLI_{2020} = 0.73) and smaller-sized rays (n = 268; RLI_{2020} = 0.79; Fig. 3A). The poor status of rays in 1970, particularly rhino rays (Rhinopristiformes), reflects steep declines and local extinctions of the larger sawfishes (Pristidae), giant guitarfishes (Glaucostegidae), and wedgefishes (Rhinidae), followed by declines in the smaller-bodied guitarfishes (Rhinobatidae; Fig. 3D). Declines of the devil rays (Mobulidae) followed, with species retained initially for their meat and subsequently also for the international trade in their gill plates that emerged in the 1980s (Fig. 3E). Declines were also observed in large stingrays (Dasyatidae), eagle rays (Myliobatidae, Aetobatidae), and angel sharks (Squatinidae) taken by target fisheries and incidental catch (bycatch), mainly of trawl and gillnet fisheries (Fig. 3F).

Much of the 14 to 32% decline in the shark RLI was driven by very steep declines in large- and medium-sized sharks (n = 37 and 75, respectively: $RLI_{1970–2020}$ = 0.94 to 0.61 and $RLI_{1970–2020}$ = 0.91 to 0.59, overall ΔRLI = 0.32), whereas small-bodied sharks declined less steeply than the overall average (n = 353, $RLI_{1970–2020}$ = 0.98 to 0.84, ΔRLI = 0.14; Fig. 3B), with much of the change due to sharp declines in hammerhead (Sphyrnidae) and requiem sharks (Carcharhinidae; Fig. 3E). Declines in the smallest size classes of sharks and rays were partially ameliorated by their higher maximum intrinsic rate of increase, r_{max} (*14*), and a decrease in predation and competition as the larger-bodied sharks were depleted (*11*, *31*).

Depletion expanding out from nearshore to surface waters of pelagic ocean

The wave of rising shark and ray extinction risk, beginning with the largest species, started in rivers, estuaries, and nearshore coastal waters before rapidly expanding out across the oceans, with lower risk only in harder-to-fish nearshore habitats and the deep sea (>200 m depth; Fig. 3, G to I). The overall deterioration of sharks and rays ranged from 11% RLI reduction in the deep sea to 36% RLI reduction in inland wetlands. Extinction risk was greatest in inland wetlands, where sharks and rays suffered an overall 36% deterioration with steep declines before 1970 ($RLI_{1970–2020}$ = 0.85 to 0.49), followed by a 34% deterioration in estuaries ($RLI_{1970–2020}$ = 0.91 to 0.57), a 29% deterioration in the intertidal zone ($RLI_{1970–2020}$ = 0.91 to 0.62); the risk continued into the fully marine continental habitats, including a 31% deterioration in coral reefs ($RLI_{1970–2020}$ = 0.95 to 0.64; Fig. 3H), a 27% deterioration in neritic sandy-mud habitat ($RLI_{1970–2020}$ = 0.95 to 0.68), and a 26% deterioration in seagrass beds ($RLI_{1970–2020}$ = 0.94 to 0.68). The RLI deterioration from nearshore to offshore reflects the local extinction of the largest, most sensitive species that are highly catchable by trawl, net, and line fisheries, for example, angel sharks (Squatinidae) and rhino rays (Glaucostegidae, Pristidae, Rhinidae, Rhinobatidae; Fig. 3, D and F). There was substantial (24%) deterioration in the pelagic ocean, which was comparable to that in the nearshore ($RLI_{1970–2020}$ = 0.95 to 0.71; Fig. 3I) owing to the early rapid expansion of pelagic longline and purse seine fisheries that began in the 1950s (*9*, *32*). The rank order of each habitat RLI changes little, with the habitats that were worst off in the 1970s progressively worsening further.

The shark and ray RLI decline is less severe in three harder-to-fish habitats: nearshore structurally complex habitats of kelp and macroalgae forests (overall 17% decline, $RLI_{1970–2020}$ = 0.99 to 0.82), neritic rocky reefs (19%, $RLI_{1970–2020}$ = 0.97 to 0.78), and the deep sea (11%, $RLI_{1970–2020}$ = 0.98 to 0.87; Fig. 3H). Deep sea fisheries tend to develop after the depletion of coastal fisheries because of the expense and technical challenges

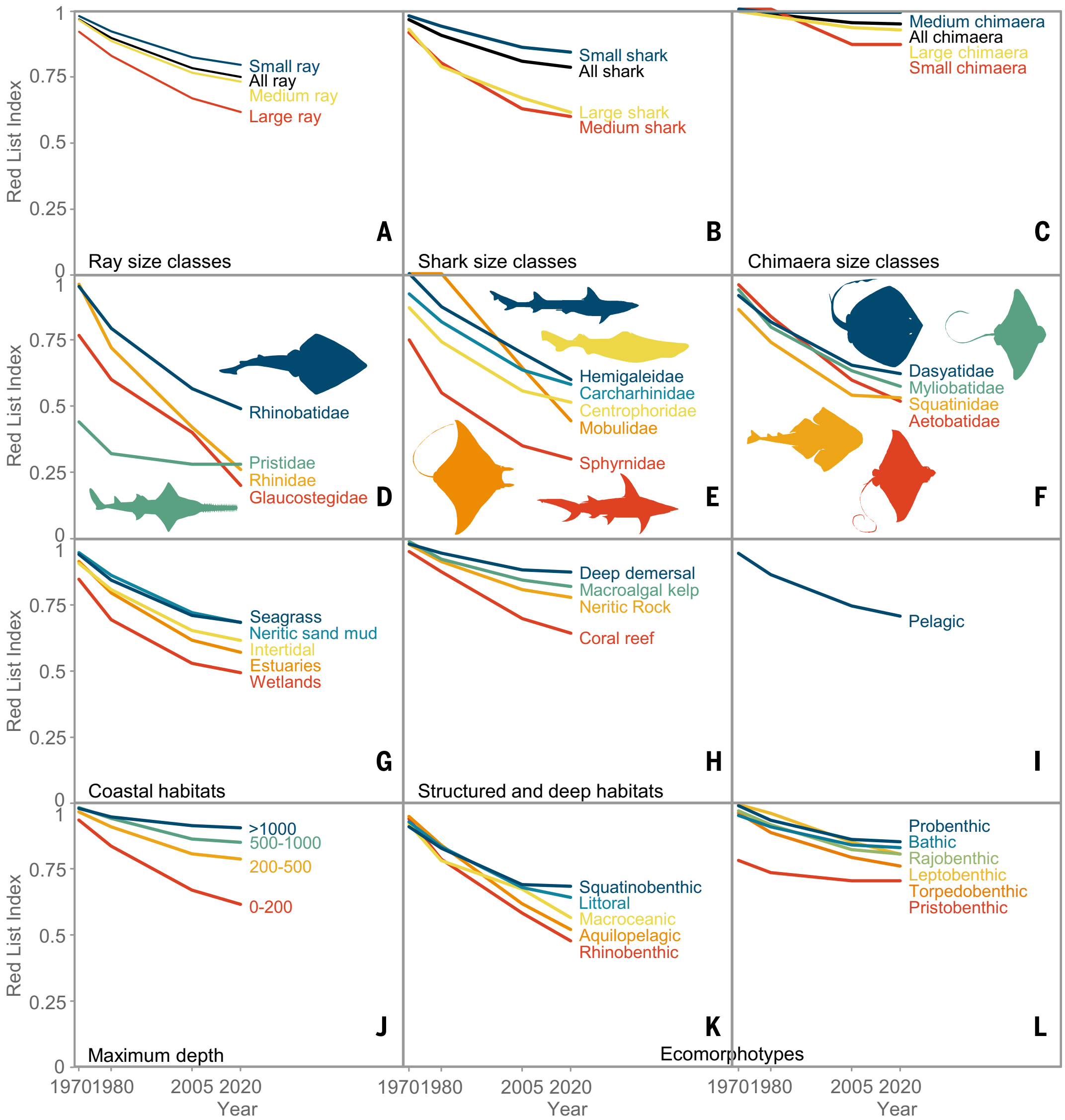

Fig. 3. Half a century of serial depletion began with the largest most sensitive species in the most accessible nearshore fishable habitats, resulting in the erosion of key families and ecomorphotypes. RLIs for the data-sufficient sharks and rays from 1970 to 2020 disaggregated by (**A** to **F**) taxonomy, (**G** to **J**) habitats, and (**K** and **L**) ecomorphotypes. (A) to (C) show RLIs by size class for each subclass of (A) rays, (B) sharks, and (C) chimaeras (see table S2 for size bins). (D) to (F) show RLIs for the 13 most threatened families (of a total of 66 families). (G) to (I) show RLIs for each major habitat, ordered left to right from (G) nearshore and continental shelf habitats that can be readily trawled with mobile fishing gear; (H) harder-to-fish habitats, including the shallow shelf biogenic coral and macroalgal and kelp reefs as well as deepwater habitats, which offer some refuge from the reach of fisheries; and (I) oceanic pelagic habitat. (J) shows RLIs by the maximum depth range of species (<200 m, 200 to 500 m, 500 to 1000 m, and ≥1000 m). (K) and (L) show RLIs by functional ecomorphotypes, with (K) showing the most at-risk ecomorphotypes, which have a RLI <0.7, and (L) showing ecomorphotypes with RLI >0.7. Diverging colors of lines and text indicate the final RLI values, with warmer colors representing low final RLI values. The monotypic archipelagic white shark *Carcharodon carcharias* was excluded from (K).

(*14*). The overall 11% reduction is substantial, given the exceedingly low population growth rates of many deepwater species, notably gulper sharks (Centrophoridae; Fig. 3E), and the rapidly expanding markets for shark liver oil (*14*). These nearshore-to-offshore habitat-specific RLI patterns can be generalized to reveal a consistent decline in RLI with depth over time, which is greatest in the shallow neritic shelf and pelagic habitat (0 to 200 m; Fig. 3J) and delayed in deeper waters (>200 m; Fig. 3J).

Erosion of ecomorphotypes and functional diversity

The half-century decreases in shark and ray population abundance and survival probability by body size class, taxonomy, habitat, and depth have resulted in the erosion of distinctive ecomorphotypes (Fig. 3, D to F). The greatest erosion occurred in the nearshore seas with the steep decline in (i) the rhinobenthic ecomorphotype composed of guitarfishes and wedgefishes (Fig. 3D); (ii) the mollusk-crushing (durophagous) eagle rays (Aetobatidae and Myliobatidae), cownose rays (Rhinopteridae), and planktivorous devil rays (Mobulidae) of the aquilopelagic ecomorphotype; and (iii) the littoral ecomorphotype, including requiem sharks (Carcharhinidae), weasel sharks (Hemigaleidae), and hammerheads (Sphyrnidae; Fig. 3, E and K), as well as squatinobenthic angel shark ambush predators (Squatinidae; Fig. 3, F and K). Steep functional declines have occurred in the iconic megapredatory macro-oceanic species that formerly dominated the pelagic ocean, notably mackerel sharks (Lamniformes), oceanic whitetip (*Carcharhinus longimanus*), silky shark (*Carcharhinus falciformis*), and whale shark (*Rhincodon typus*; Fig. 3, E, I, and K). By comparison, the functional groups that have declined less include deepwater bathic dogfishes (Squaliformes), leptobenthic and probathic catsharks (Pentanchidae, Scyliorhinidae, Proscyliidae), rajobathic skates (Rajidae, Gurgesiellidae, Arhynchobatidae, and Anacanthobatidae), and torpedobenthic electric rays (Narkidae, Hypnidae, Torpedinidae; Fig. 3L). Ecosystems may have redundancy because multiple species, often from the same family, possess similar functional traits. Alternatively, the clustered nature of deteriorations by body size, habitat, and ecomorphotype could result in a rapid loss of ecological function and evolutionary distinctiveness because whole families are sensitive to overfishing (*12*, *20*, *33*).

To test this hypothesis, we evaluated the global erosion of ecological function by measuring the

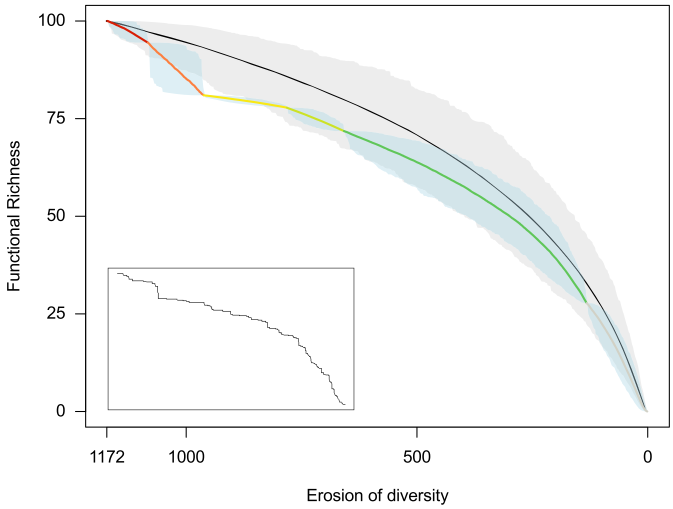

Fig. 4. Overfishing-driven extinction risk sequence driving disproportionate loss of ecological function of sharks and rays. The loss of functional richness through the 500 bootstrapped simulated extinction trajectories of 1172 shark and ray species sequenced from the most to least threatened IUCN Red List categories: Critically Endangered (possibly extinct; dark red), Critically Endangered (red), Endangered (orange), Vulnerable (yellow), Near Threatened (light green), Least Concern (dark green), and Data Deficient (gray) compared with random loss of species (black), with the range of possibilities encompassed within the gray polygon (95% CI). Although the overfishing-driven sequences of extinction risk (colored lines and polygons) lie close to the random sequences of extinction risk (black line and gray polygon), they lie below and outside of the range of random chance possibilities. The inset is a single simulation showing the nonlinear and highly disproportionate loss of functional richness that results from the elimination of a species on the vertex, defining the boundary of the multidimensional functional diversity volume (figs. S6 and S7). These simulations were insensitive to alternate classifications of Data Deficient species (fig. S8).

reduction in the functional trait volume occupied by species, as measured by the functional richness index (10, 34). We simulated the erosion of functional richness according to the recent pattern of species extinction risk by eliminating the most threatened species first and the Least Concern and Data Deficient species last. The present sequence of rising extinction risk due to serial depletion is eliminating functional diversity faster than would be expected by chance alone (Fig. 4). The variability in the decline of functional diversity is due to the nonlinear stepwise changes in individual extinction trajectories attributable to the simulated extinction of species at the edge of the functional trait volume (inset of Fig. 4). The elimination of Critically Endangered and Endangered species would cause a disproportionate erosion (19%) of functional diversity, which is 12% greater than random extinction (7%), and the extinction of threatened species would result in an overall 22% erosion of functional diversity, which would be an 8% greater loss than expected under random extinction scenarios (gray polygon in Fig. 4). This is because the clustered extinction risk of the largest size classes, species, and families (Fig. 3, A to F) has greater functional value (Fig. 3, G to L), and our finding is consistent with the projected loss of 44 to 87% in functional richness (11 to 18% greater than expected by chance) in 71 megafaunal sharks and rays (10).

Bright and dark spots in the shark and ray RLI

The spatial patterning of the loss of species tracked by the change in RLI over the past 50 years is closely related to the present distribution of threatened species. The places with the greatest percentage of threatened species are also the places with the greatest change in threat, as indexed by the range-weighted RLI (fig. S3). The greatest broad-scale increase in shark and ray extinction risk occurred in the Southwest Atlantic, Eastern Central Atlantic, and Mediterranean and Black Seas (Fig. 5A). Focusing on large marine ecosystems, the greatest increase in extinction risk occurred in the highly diverse ecosystems of the (i) tropical Atlantic Ocean, (ii) Northern Indian Ocean, (iii) Indo-west Pacific Ocean (specifically in the highly diverse Gulf of Thailand), and (iv) the Mediterranean Sea and Black Sea regions (Fig. 5B). Fine-scale increasing extinction risk is apparent in the ecoregions of the Northeast Atlantic and Western Indian Ocean (Fig. 5C). At the national scale, we summarized the change in RLI relative to species richness, focusing on bright spots of high species richness with relatively little reduction in RLI (green to yellow in Fig. 5D) in contrast to dark spots of high richness and RLI reduction (purple to blue in Fig. 5D).

Bright spots include countries that have some science-based fishery management measures that benefit sharks and rays, with the aim of long-term sustainability (16, 35) primarily: (i) the United States and Canada; (ii) the Northeast Atlantic west of Scotland, Ireland, France, Spain, and Portugal; (iii) the southwestern cape of South Africa; and (iv) Australia and New Zealand (green to yellow in Fig. 5D). We caution that these bright spots have at least three challenges, including cases of serious depletion and/or ongoing unsustainable mortality for many populations (36), fleets operating unsustainably elsewhere on the high seas or in other countries' EEZs (37), and/or the import of seafood from inequitable fisheries elsewhere (38).

The dark spots of the highest species richness and greatest RLI decline occur in areas with weak fisheries management spanning five main tropical coastal regions (purple to blue in Fig. 5D), including places of high functional richness and evolutionary distinctiveness (13, 17, 33). The whole tropical Indo-Western Pacific Ocean ranging from (i) the Southwestern Indian Ocean nexus of Kenya, Tanzania, Mozambique, Madagascar, and surrounding insular nations to (ii) the Arabian Sea and adjacent waters (the Red Sea and the Persian/Arabian Gulf) and across the Northern Indian Ocean to New Guinea and Korea in the west Pacific Ocean; (iii) West Africa (Morocco to Angola) in the East Atlantic Ocean and (iv) the Mediterranean and Black Seas; (v) Brazil to northern Argentina in the Southwest Atlantic; and (vi) Mexico to Ecuador in the Eastern Pacific (purple to blue in Fig. 5D). By comparison, the pelagic ocean has much lower diversity, with the greatest reduction in RLI adjacent to tropical and subtropical shelves and major archipelagoes and notably across much of the North Atlantic Ocean and adjacent to either western Indian Ocean archipelagos or Indo-West Pacific continental shelves (purple in Fig. 5D).

These dark spots also include places of high endemicity (range-weighted rarity) and threat (range-weighted RLI), notably (i) Japan to Taiwan; (ii) Indonesia to New Guinea; (iii) India and Bay of Bengal; (iv) Western Indian Ocean to South Africa; (v) Brazil, Uruguay, and Argentina; (vi) Galapagos and northwest South America; and (vii) the Gulf of California (fig. S5). There were also two areas of lower threat and high endemicity: (viii) east and west Australia and (ix) the Bahamas, Florida, and Hispaniola hotspot of deepwater endemicity (fig. S5).

Fishing pressure and management capacity influence national extinction risk

National shark and ray extinction risk (RLI; Fig. 6A) is influenced by nine variables organized into three classes: (i) ecological carrying capacity (area of EEZ, sea surface temperature, and primary production), (ii) fishing pressure (CPUE, coastal human population size, and harmful fisheries subsidies), and (iii) management

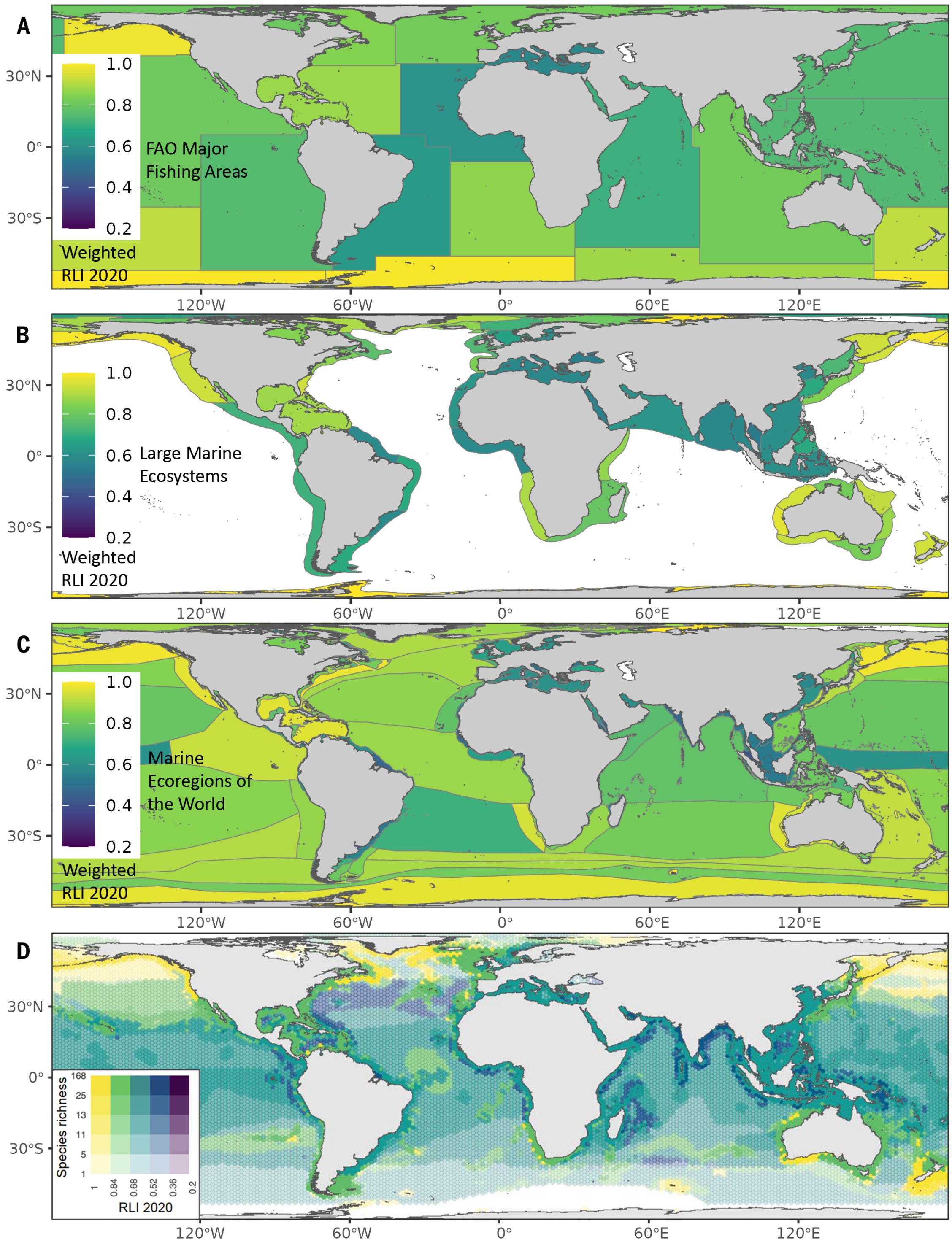

Fig. 5. Bright and dark spots of the global rarity-weighted RLIs of shark and ray values across in 2020 at four biogeographic scales. (**A**) Major fishing areas of the Food and Agriculture Organization of the United Nations (FAO). (**B**) Large marine ecosystems. (**C**) Marine ecoregions of the world. (**D**) Scale of individual hexagons (~23,000 km^2).

capacity [world governance index (WGI), gross domestic product (GDP), and beneficial fisheries subsidies]. These three variable classes account for 40.5, 30.8, and 28.7% of the summed variable importance from 1000 bootstrapped boosted regression trees, respectively (Fig. 6K) (cross-validated prediction bias is equal to −0.0018 of an RLI value, and the bootstrapped range is equal to −0.075 to 0.058; Fig. 6 and table S4). Considering the ecological carrying capacity variables first, habitat availability (i.e., EEZ area) best predicted national RLI values: Larger EEZ areas corresponded to countries with lower extinction risk (relative variable importance equal to 17.0%; Fig. 6B). Extinction risk tended to be higher in EEZs with warmer sea surface temperatures (12.7%; Fig. 6C) and lower primary production (10.8%; Fig. 6D), consistent with

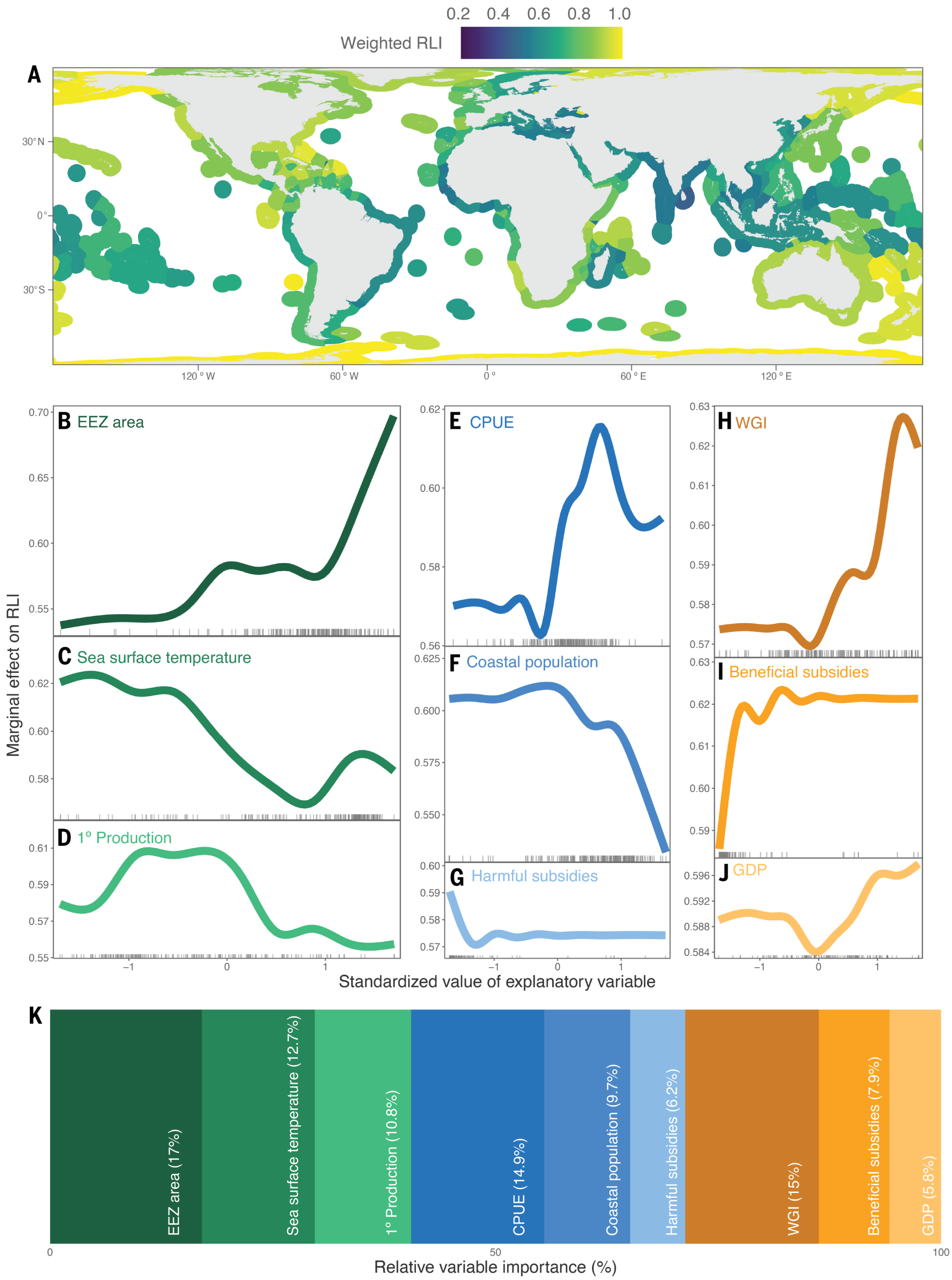

Fig. 6. Ecological carrying capacity and the role of fishing pressure and management capacity on the national RLI. (**A**) National weighted RLI in each EEZ. (**B** to **J**) Marginal effect on RLI of [(B) to (D)] ecological carrying capacity (green), [(E) to (G)] fishing pressure (blue), and [(H) to (J)] management capacity (orange) ordered by variable importance. Darkest colors and larger panels indicate variables with the highest importance, with lighter colors and smaller panels indicating the opposite. Note that the *x* axes have been scaled and centered for presentation purposes, whereas the raw numbers were used in the analysis. The rug at the bottom of each panel denotes the distribution of raw data points from the model. Explanatory variables include (B) EEZ area, (C) sea surface temperature, (D) primary production, (E) CPUE, (F) coastal population, (G) harmful fishery subsidies, (H) WGI, (I) beneficial fishery subsidies, and (J) GDP. (**K**) The relative variable importance of each variable. Larger boxes indicate higher importance and are colored based on the colors used in (B) to (J).

extinction risk being greater in the tropics (Fig. 5, B to D). Extinction risk was greater in countries with undermanaged fisheries and high fishing pressure, driven by lower CPUE of sharks and rays (14.9%; Fig. 6E), larger coastal human population sizes (9.7%; Fig. 6F), and larger contributions of harmful "bad" fisheries subsidies (6.2%; Fig. 6G). Fishing pressure is underlain by the macroeconomic drivers of sustainability, such that countries with lower extinction risk (higher RLI values) also had stronger WGI values (15.0%; Fig. 6H); greater contributions towards beneficial "good" fishery subsidies, which are directed toward scientific-based catch control (7.9%; Fig. 6I); and higher GDP (5.8%; Fig. 6J).

Discussion

The past half-century has seen rising concern over the scale of biodiversity extraction from our oceans (*1*, *2*), which has reached a state of emergency according to the UN (*3*). In this work, we quantified the global fishing impacts on biodiversity and ecosystem function of the greatest radiation of marine predators. The global consequences of underregulated fisheries and international trade (*1*) have elevated shark and ray extinction risk across four dimensions of body size, ecomorphology and function, habitat, and geography.

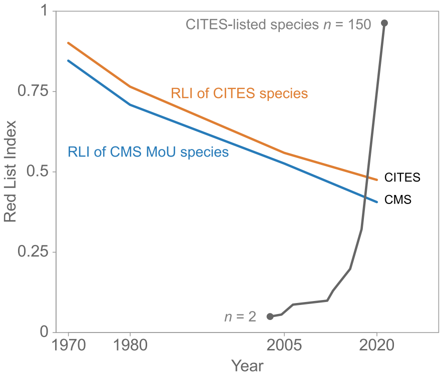

Fig. 7. The rapid growth in the number of species listed in the appendices of international conservation treaties and the RLIs to indicate their conservation dependence. RLIs for sharks and rays listed in appendices I and II of CITES (orange) and the Convention on Migratory Species (CMS) Memorandum of Understanding (MoU) for Sharks appendices (blue). The gray line shows the number of species listed in CITES appendices I and II from 2003 to 2023 (table S1).

The geographic expansion of fisheries outward from freshwaters and estuaries into the nearshore and surface waters of the ocean was driven by the rapid expansion in food demand from growing coastal populations and associated artisanal and subsistence fisheries, as well as the expansion of large-scale fisheries (Fig. 3, G to I). Dwindling coastal fishery yields and rising harmful fisheries subsidies drove a bathymetric expansion of fisheries down the deep slopes and out to seamounts (Fig. 3J). This pattern of fisheries expansion and technological intensification increased extinction risk, with steep population declines and widespread elimination of the larger-bodied, functionally distinct species. The lost shark and Javan stingaree may well have already disappeared unnoticed before 1970 owing to coastal fisheries. Further, the larger rhino rays (Rhinopristiformes), such as sawfishes, were depleted in many places before 1970. The large, accessible basking shark (*Cetorhinus maximus*) and Greenland shark (*Somniosus microcephalus*), valued for centuries for their rich liver oil, were depleted by the 1900s and are now largely protected from targeted fishing (*14*). These declines were accompanied by a sequential depletion of other large valuable macropredatory species, leaving smaller mesopredatory species (Fig. 3, A to J), which is particularly well documented on coral reefs (*11*, *31*). The expansion of tuna and billfish fisheries into the open ocean drove steep declines in pelagic sharks, especially the evolutionarily and functionally distinct lamnids and hammerheads in the Atlantic Ocean (*9*, *13*, *32*, *33*) (Fig. 3E). This serial depletion, beginning with the largest species, has led to widespread ecological erosion of functionally distinct morphotypes (Figs. 3K and 4A), shaping the spatial patterning of functional richness, distinctiveness, and specialization (*33*). Although declines were initiated by overfishing, other threats—including coastal habitat degradation, climate change, sublethal effects of pollutants, and ship strikes—exacerbate overexploitation and hinder recovery (*12*, *20*).

Although serious gaps remain, there has been notable progress in the appreciation and conservation of sharks and rays over the past 35 years, including the 1991 Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) Shark Resolution, the 1999 International Plan of Action for Conservation and Management of Sharks, the first CITES listings for sharks in the 2000s, the first national shark protections of the 1990s, the global wave of finning bans in the 2000s, the first regional fishery management organization shark limits in the 2000s, and the first listings for commercially exploited sharks under CITES in 2013 (*16*, *18*, *39*). Although there are many factors underlying progress, we note the increasing acceptance of the validity of Red List assessments by fisheries scientists and managers (*9*), increasing (albeit too slowly) collaboration between fisheries and wildlife treaty authorities, concerted development of scientific advice, and consequential advocacy by conservation nongovernmental organizations (*16*, *18*, *39*). At present, 150 shark and ray species are listed in appendix I or II of CITES, encompassing more than 85% of the global fin trade volume and all the gill plate trade (gray ascending line in Fig. 7). Although these listings are driving some national fisheries assessment and conservation measures, compliance, monitoring, and enforcement remain formidable challenges. Chronic underassessment and undermanagement of shark and ray fisheries persist (*1*, *17*, *18*), as does a lack of implementation of other conservation treaties (*16*, *39*). Countries can have plenty of fisheries management "capacity" but still deplete sharks and rays because of the lack of political will to restrict overfishing (*16*).

Science-based catch limits and measures to minimize incidental mortality are not only essential to meeting CITES obligations for listed sharks and rays but also fundamental for overall sustainability of all exploited species (regardless of CITES listing). Myriad restrictions are urgently needed around the world to restore depleted species' ecological, social, and economic potential. Spatial protections also have considerable potential to abate fishing mortality, especially if they are well tailored to the biology of a species and its specific threats (*16*, *36*, *40*).

Further erosion of biodiversity and ecosystem function would be even more detrimental to humanity because more than a billion people around the world rely on oceans for nutrition and livelihoods (*2*). Protecting aquatic species, ecosystems, and their foundational contributions to sustainable development has been a long-standing priority of the UN. Numerous pledges and declarations have gone woefully and quietly unmet in part because of the lack of availability of indicators of the status of the wider range of species killed by fisheries, particularly sharks and rays (*8*). The IUCN Red List assessment approach and the RLI are strongly aligned with stock assessments and are complementary in tracking the status of unassessed and sparsely managed species (*6*, *9*). We recommend the adoption of a RLI for sharks and rays as an indicator for SDG 14.4 (disaggregated for the impacts of fisheries) and for KMGBF target 5 (disaggregated for the impacts of harvest), complementing the existing fisheries stock assessment indicator (*6*). RLIs offer a step toward greater alignment between environment and fisheries policy realms, with a complementary view of the status and trends of the wider range of exploited fishes and fisheries that underpin the ecosystem functions and services that support the sustainable development of humanity.

Materials and methods

Global chondrichthyan catch and effort

We calculated annual CPUE by dividing reconstructed catch for each EEZ (1950–2019) by fishing effort of each nation, which required projecting effort from 1950 to 2010 out to 2019 using a Bayesian random walk model.

Red List assessments and the calculation of RLIs

We considered the 1199 mainly marine species of class Chondrichthyes (sharks, rays, and chimaeras, hereafter "sharks and rays") assessed up to 31 March 2021 using the IUCN Red List Categories and Criteria (version 3.1) to classify species into six of the eight IUCN Red List categories: Data Deficient, Least Concern, Near Threatened, Vulnerable, Endangered, or Critically Endangered (*20*). A RLI was calculated based on the number of species in each Red List category at each of four time periods over a half-century (1970, 1980, 2005, and 2020). The 2020 RLI was derived from the recent reassessment of all species (spanning 2012 to 2021) (*20*). The 2005 RLI assessment was based on the first global assessment (1996 to 2011) (*19*) with species' categories retrospectively corrected to account for nongenuine changes, which result from taxonomic changes, new knowledge, and improvements in the application of criteria (*4*, *5*, *20*). We calculated RLIs for different subsets of 1044 data-sufficient sharks and rays, including by taxonomy, maximum body size and other ecological traits, maximum depth, IUCN level 2 habitat, major ecomorphotype, and spatial units. We used a bootstrap method to quantify the effect of two uncertainties on the RLI: of classifying Data Deficient species as either Least Concern or Critically

Endangered, and systematic classification error by assessors (fig. S1).

RLIs were mapped based on IUCN Red List species distribution maps (*20*), with an RLI calculated for each time slice for each of five spatial layers: (i) major fishing area, (ii) large marine ecosystem, (iii) marine ecoregion of the world, (iv) EEZ, and (v) each individual hexagonal cell (~23,322 km^2). The contribution of each species to the RLI was the product of each species' IUCN Red List category weight [0 for Least Concern through 4 for Critically Endangered and Critically Endangered (possibly extinct)] and the fraction of its geographic range within the spatial unit. The final RLI value in 2020 is a result of the magnitude of change since 1970 (fig. S3). The range of possible RLI values scales with species richness, and the variability in values is funnel shaped, with greatest variance at low richness values (fig. S4). Consequently, we defined bright spots and dark spots based on the combination of species richness and the RLI in 2020, reasoning that poor (low) RLI values in species-rich places (dark spots) are of greater conservation concern than in low-richness places. We calculated rarity-weighted species richness based on the species list in each unit weighted by the fraction of range in each spatial unit (fig. S5).

Simulations of the loss of functional diversity

We calculated the potential ecological erosion of sharks and rays based on present extinction risk by simulating changes in functional richness calculated from four functional traits: maximum body size, geographic range, a three-level categorization of reproductive mode, and maternal investment strategy reflecting the fast-slow physiology and life history nexus, as well as 35 IUCN habitat classes (figs. S6 and S7 and table S3). We calculated functional richness as the minimum convex hull of a principal coordinates analysis space based on a matrix of Gower pairwise species distances. We estimated the effect of the present extinction trajectory by removing species one by one on the basis of their IUCN Red List category (random draws from Critically Endangered species first to Data Deficient species last) and recalculating functional richness, then contrasting this to a scenario of species removal at random. We also evaluated sensitivity to data deficiency by assuming that Data Deficient species were either Least Concern or Critically Endangered (fig. S8).

Modeling national coastal extinction risk at the EEZ scale

We modeled the nation-scale RLI of coastal sharks and rays using boosted regression tree analyses of nine national-level covariates spanning three classes: ecological carrying capacity, fishing pressure, and management capacity (table S4). We used a cross-validation approach by randomly separating the data into an 80% training–20% test set to build and validate models. We measured the model's performance by calculating a prediction bias by subtracting the predicted RLI value from the observed value, whereby a bias close to zero indicates a well-predicting model (fig. S11). For more details, see the methods section in the supplementary materials.

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SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.adn1477
Materials and Methods
Figs. S1 to S11
Tables S1 to S4
References (*41*–*146*)
MDAR Reproducibility Checklist

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